LeoWay — Project Proposal

Team: LRD (Leo Research Department)

Authors: Laurenz (LPuehringerStudent), Ayan (Ayan2310), David (David-Fruehwirt), Behrang (BehrangEl11)

Summary

We propose LeoWay, a commute assistant for HTL Leonding students. It combines each student's WebUntis timetable with live public transport data. It tells students when to leave for their actual first lesson and warns them early about delays. When a delay makes a student late, LeoWay drafts an appeal to the form teacher with a delay record the teacher can check. The project needs no paid licences, runs over two school years in two-week sprints, and its core is already shown to work with HTL Leonding's WebUntis.

1. Initial situation

Most HTL Leonding students come to school by public transport. The school is served by Linz AG bus 19 at Limesstraße, by trams 3 and 4 and buses 11, 19 and 191 at Meixnerkreuzung, and by regional OÖVV buses and ÖBB trains. Lessons start at 07:05 or 08:00, and the school plans them in WebUntis, including cancellations, substitutions and exams.

How it works today

1. In the evening or morning, the student checks WebUntis for changes to the first lesson.
2. Separately, the student checks Scotty or the Linz AG app for departures and works out when to leave.
3. A delay is usually noticed at the stop, when an earlier connection is already gone.
4. If the student arrives late, the teacher records a lateness entry in WebUntis.
5. To excuse it, the student sends the form teacher a photo of the vehicle display or a Scotty screenshot.
6. The form teacher decides, but cannot check whether the picture is genuine or from that morning.

Deficits

- **No connection between timetable and transport.** Transit apps don't know when the first lesson starts or that it was cancelled. WebUntis doesn't know when the bus leaves.

- **Warnings come too late.** Nobody warns the student while there is still time to take another connection.
- **Evidence can't be trusted.** Photos and screenshots are easy to fake or reuse, so teachers decide on weak evidence and may decline genuine cases.
- **No data.** Nobody records which connections make students late, so the student council has nothing to take to Linz AG or OÖVV.

The gap. All the data already exists: WebUntis has the timetable and lateness entries, and Linz AG publishes live delays. What is missing is a system that connects them for each student, warns in time and keeps a record that teachers can trust.

2. General conditions and constraints

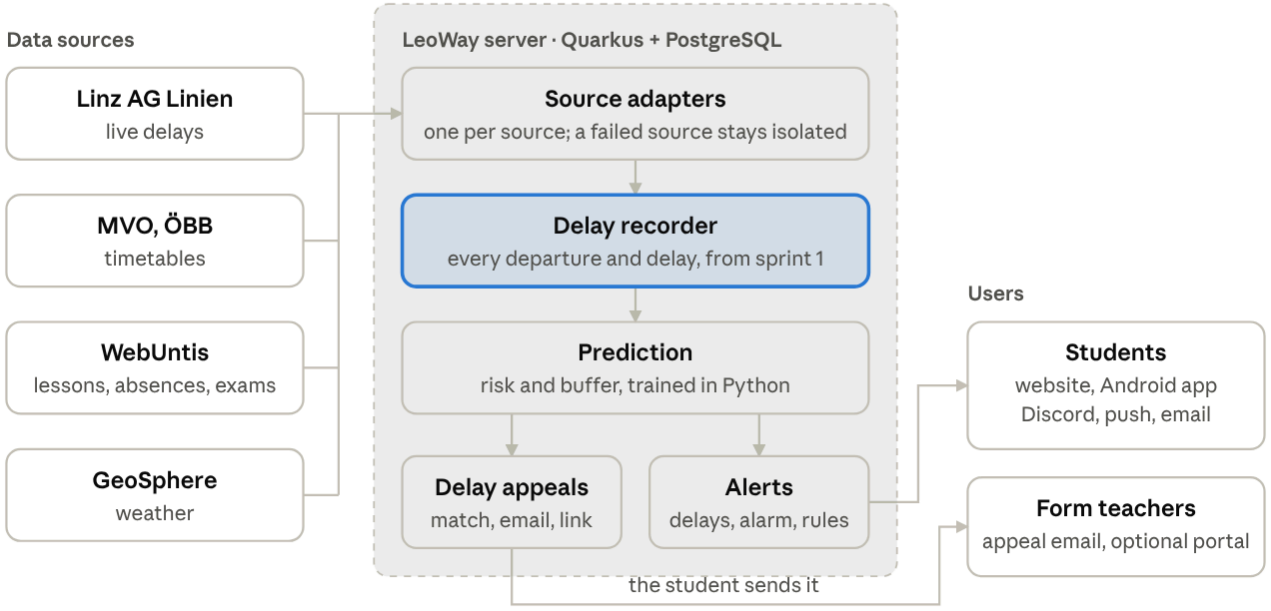
Framework conditions

- **Budget:** none. LeoWay is a school project, so it must run on free services, free data and the team's own hardware.
- **Organisation:** a team of four in the SYP course, supervised by our teacher, who acts as the customer. We work in Scrum with two-week sprints over two school years.
- **Know-how:** the team knows Java, TypeScript and SQL from school. Quarkus, Angular and PostgreSQL are taught at HTL Leonding; Capacitor, GTFS feeds and model training are new and planned into early sprints.
- **Partners:** school IT (permission to use WebUntis access keys), form teachers (what an appeal must show), Linz AG and VAO (transport data), and the student council (statistics).
- **Deadlines:** the project follows the school year; milestones are in section 5.

Technical conditions

The backend runs on Quarkus with a PostgreSQL database, the website on Angular, and the Android app is the same Angular code wrapped with Capacitor, because a website can't ring an exact-time alarm. The prediction model is trained in Python on our own hardware. The server runs on the team's server or LeoCloud, the school's hosting.

Every feature builds on the recorded delay history



LeoWay architecture · 4 sources, 5 components, 2 user groups

Source	Data	Access
Linz AG Linien	Live departures and delays	Free
MVO, ÖBB	Bus and train timetables (GTFS)	Free; ÖBB under CC BY 4.0
WebUntis	Timetable, exams, lateness entries	The student's own access key, read only
GeoSphere Austria	Weather, for prediction	Free
VAO Start	Journeys across Austria	Free tier only 100 queries a day; education licence to be requested

We have already tested HTL Leonding's WebUntis with a real student account: login with the access key takes about 0.6 seconds without a password, and the timetable, cancellations, exams and lateness entries all load.

Legal constraints

- **GDPR and the Austrian Data Protection Act** apply. Use is voluntary, and students aged 14 and over can consent themselves.

- **Data minimisation:** no GPS, no home address and no WebUntis password. The access key is stored encrypted; absence reasons and notes are never stored.
- **Retention:** appeals and lateness entries are deleted at the end of the school year or with the account.
- **School rules:** LeoWay only reads WebUntis and never excuses anything; the form teacher still decides.
- **Data licences:** ÖBB data is credited in the app; VAO is used only within its contract.

3. Project objectives and system concept

The idea

The student connects WebUntis once and saves a route: home stop, school stop and walking times. From then on, LeoWay does the thinking every school day:

- **In the evening,** it reads tomorrow's first real lesson and sets the leave-by time. If the first lesson is cancelled, the time moves, and on Android so does the alarm.
- **In the morning,** it watches the student's bus or tram. If a delay will make the student late, it warns them in Discord, by push or email, and suggests another connection. If the first lesson is an exam, the warning says so.
- **On the way,** students can report a disruption in a few taps. When three students report the same one, everyone on that line is warned, often before official feeds show it.
- **After a late arrival,** LeoWay finds the lateness entry in WebUntis and matches it with the recorded delay: "You were marked 13 min late; your bus 19 was 11 min late." One tap opens an email to the form teacher with the journey details and a link. The teacher opens the link and sees the delay record, not a screenshot.
- **Over time,** the recorded delays show how risky tomorrow's connection is, suggest a personal buffer, and tell the student council which connections regularly arrive after lessons start.

LeoWay only reads WebUntis; the student sends the email and the teacher still decides. Teachers who want to can handle appeals in an optional portal instead of by email.

Objectives

Objective	Target
Timetable-aware alerts	Every alert uses the student's first real lesson, including cancellations, substitutions and exams

Objective	Target
Early warnings	A student whose connection will arrive late is warned and offered another one where it exists
Trusted appeals	A late student prepares an appeal in under one minute; the teacher checks it through the link
Data history	Delays are recorded from the first sprint, including holidays
Crowd reports	Three reports of the same disruption warn everyone on that line
Prediction	The model beats "same delay as yesterday" and "average delay of this trip"
School statistics	One report on late connections, handed to the student council

Not part of the project: writing to WebUntis, location tracking, a native iPhone app (iPhone users get the website) and proving that a student was on the vehicle.

4. Opportunities and risks

Market

LeoWay is free and non-commercial, so its "market" is the people who gain from it:

User	What they gain
Students who commute by bus, tram or train	Leave on time, fewer unexcused late arrivals, less stress on exam days
Form teachers	Appeals they can verify in seconds instead of judging screenshots
The school and student council	Evidence about connections that make students late
Later: other schools in Linz	The same benefit; WebUntis is widely used in Austrian schools

Existing apps cover only one side. Scotty and the Linz AG app know departures but not the timetable; the WebUntis app knows the timetable but not the bus. We know of no app that connects the two or produces evidence for lateness appeals.

Opportunities

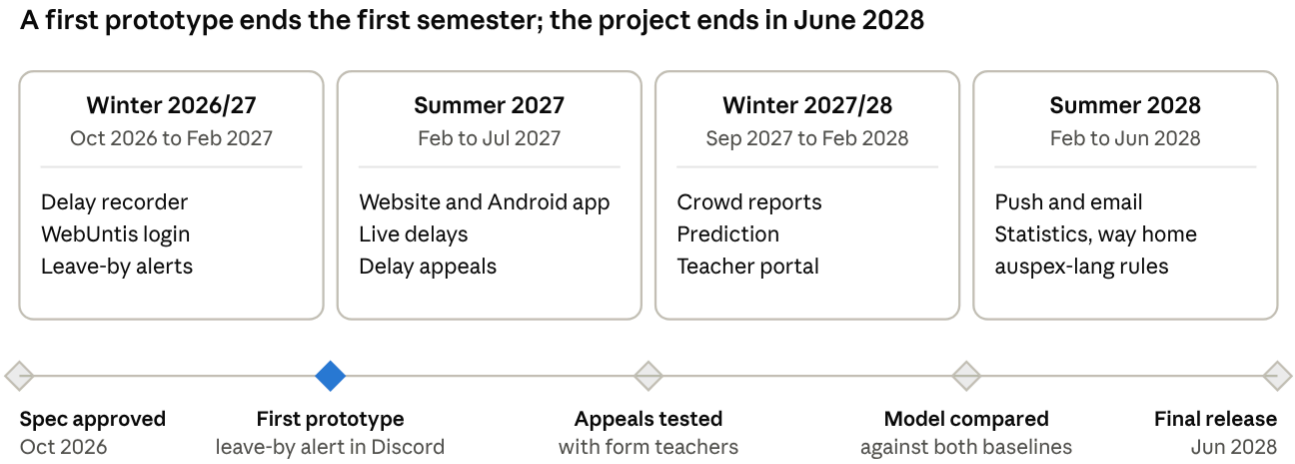
- Students arrive on time more often, because they are warned while there is still a choice.
- Teachers spend less time on lateness excuses and decide on real data.
- After one school year, LeoWay holds a delay history for the lines around the school that no one else collects.
- The design works for any WebUntis school, so it can grow to other schools in Linz.

Risks

Risk	Effect	Fallback
The school or Untis blocks access keys or the WebUntis API	No automatic timetable	iCal link or manual entry behind the same adapter
No VAO education licence	No live data for regional buses and trains	Live data for Linz AG lines, timetables for the rest
A feed goes down or changes format	One source missing	Cached data, a "timetable only" notice, an alert to the team
The model doesn't beat the baselines	Weaker risk scores	The baselines run anyway; the comparison is a result
Android delays exact-time alarms	Alarm rings late	Tested early; a high-priority notification instead
Teachers don't accept appeals	Appeal feature unused	Agree the format with form teachers before building it
Few students use LeoWay	Little feedback	Start with our own classes; the Discord bot needs no install

5. Planning

Milestones



Planned milestones · 4 semesters, 5 gates

The project starts with this proposal and the system specification in October 2026; implementation starts in the first sprint after the specification is approved. The delay recorder comes first so the history starts early. A first prototype, a leave-by alert in Discord based on the real timetable, is planned for the end of the first semester. The project ends with the final release in June 2028. Each phase takes the next epics from the backlog, and the order is reviewed in every sprint review.

Roles

Member	Role
Laurenz	Project lead and developer; runs the GitHub repository, assigns issues, reviews and merges pull requests
Ayan	Scrum master and developer
David	Developer
Behrang	Developer
Supervising teacher	Customer
Product owner	Open, to be found before the first sprint

Work is assigned per sprint, so every developer works on backend, apps and data. The definition of done: code reviewed, tests passing in CI, deployed to the test server and documented.

Resources

- **People:** the four team members, plus students from our classes and at least one form teacher in sprint reviews.
- **Hardware:** a test server (team server or LeoCloud), one Android phone for alarm tests, and our own computers for model training.
- **Licences and access:** all software is open source; WebUntis permission from school IT; a VAO education licence is requested but not required.

Before the first sprint

- [ ] Ask school IT whether students may use their WebUntis access key, and how often we may sync
- [ ] Ask one or two form teachers what an appeal must show for them to accept it
- [ ] Apply for a VAO Start education licence
- [ ] Test whether the Capacitor app can ring an exact-time alarm on Android
- [ ] Find a product owner and decide on hosting

6. Economic efficiency

LeoWay costs almost no money; the main cost is the team's working time.

Item	Cost
Software (Quarkus, Angular, PostgreSQL, Capacitor, Python)	Free, open source
Transport and weather data (Linz AG, MVO, ÖBB, GeoSphere)	Free
WebUntis access	Free, with each student's own key
Hosting	Free on LeoCloud or the team's own server
Discord bot, push notifications, email drafts	Free
Model training	Team's own hardware
Google Play developer account	US$25 once, only if the app is published there instead of installed directly

Item	Cost
VAO education licence	Open; VAO offers education discounts, and LeoWay works without it

Benefit. Students lose less time and stress to late arrivals, form teachers decide appeals faster and on real data, and the school gets evidence it can use with Linz AG and OÖVV. None of this has a running cost for the school.